Sommersemester 2026

6th Exercise Sheet

TUTORIUM: Friday, 26.06.2026.

Consider the Hubbard-Hamiltonian given by

$$\hat{H} = \sum_{\boldsymbol{k}\sigma} \varepsilon_{\boldsymbol{k}} \hat{c}^\dagger_{\boldsymbol{k}\sigma} \hat{c}_{\boldsymbol{k}\sigma} + U \sum_i \underbrace{\hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow}}_{\hat{n}_{i\uparrow}} \underbrace{\hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}}_{\hat{n}_{i\downarrow}} - \mu \sum_i \underbrace{(n_{i\uparrow} + n_{i\downarrow})}_{\hat{n}_i}. \tag{1}$$

The term containing μ fixes the number of particles. Specifically, we consider the case of half-filling where we have $\langle \hat{n}_i \rangle = 1$ particle per site. This corresponds to $\mu = \frac{U}{2}$.

SECTION

012

The Green's function in limiting cases

PROBLEM

First, assume that the electrons governed by the Hamiltonian in Equation 1 are non-interacting, i.e., $U = 0$.

a)

PROBLEM

Compute the one-particle Green's function $G_\sigma(\boldsymbol{k}, \tau)$ by directly calculating the trace in the definition

$$G_\sigma(\boldsymbol{k}, \tau) = -\frac{1}{Z} \operatorname{tr}\left[\mathrm{e}^{-\beta \hat{H}} \hat{c}_{\boldsymbol{k}\sigma}(\tau) \hat{c}^\dagger_{\boldsymbol{k}\sigma} \right], \quad \beta \geq \tau \geq 0, \tag{2}$$

of the Green's function. The partition function is defined as $Z = \operatorname{tr}\left[\mathrm{e}^{-\beta\hat{H}}\right]$.

Hint: Use the Lehmann representation, i.e. perform the trace over the basis of the eigenvalues and insert the completeness relation, where needed.

SOLUTION

The noninteracting $U = 0$ Hamiltonian

$$H = \sum_{\boldsymbol{k}\sigma} \xi_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}\sigma} c_{\boldsymbol{k}\sigma} \qquad \xi_{\boldsymbol{k}} = \varepsilon_{\boldsymbol{k}} - \mu \tag{3}$$

has eigenstates

$$|\{n_{\boldsymbol{k}\sigma}\}\rangle \qquad n_{\boldsymbol{k}\sigma} \in \{0, 1\} \tag{4}$$

with eigenenergies

$$E_{\{n\}} = \sum_{\boldsymbol{k}\sigma} \xi_{\boldsymbol{k}} n_{\boldsymbol{k}\sigma}. \tag{5}$$

The partition function is then given by

$$Z = \sum_{\{n\}} \mathrm{e}^{-\beta E_{\{n\}}} = \prod_{\boldsymbol{k}\sigma} \sum_{n_{\boldsymbol{k}\sigma}=0}^{1} \mathrm{e}^{-\beta \xi_{\boldsymbol{k}} n_{\boldsymbol{k}\sigma}} = \prod_{\boldsymbol{k}\sigma} \left(1 + \mathrm{e}^{-\beta \xi_{\boldsymbol{k}}}\right). \tag{6}$$

Green's function:

$$G_\sigma(\boldsymbol{k}, \tau) = -\frac{1}{Z} \sum_{\{n\}} \left\langle \{n\} \middle| \mathrm{e}^{-\beta H} c_{\boldsymbol{k}\sigma}(\tau) c^\dagger_{\boldsymbol{k}\sigma} \middle| \{n\} \right\rangle \tag{7}$$

Because $|\{n\}\rangle$ is an eigenstate of H,

$$G_\sigma(\boldsymbol{k}, \tau) = -\frac{1}{Z} \sum_{\{n\}} \mathrm{e}^{-\beta E_{\{n\}}} \left\langle \{n\} \middle| c_{\boldsymbol{k}\sigma}(\tau) c^\dagger_{\boldsymbol{k}\sigma} \middle| \{n\} \right\rangle. \tag{8}$$

Inserting a resolution of unity

$$1 = \sum_{\{m\}} |\{m\}\rangle \langle \{m\}| \tag{9}$$

we get

$$G_\sigma(\boldsymbol{k}, \tau) = -\frac{1}{Z} \sum_{\{n\},\{m\}} \mathrm{e}^{-\beta E_{\{n\}}} \langle \{n\} | c_{\boldsymbol{k}\sigma}(\tau) | \{m\} \rangle \left\langle \{m\} \middle| c^\dagger_{\boldsymbol{k}\sigma} \middle| \{n\} \right\rangle. \tag{10}$$

Using

$$c_{\boldsymbol{k}\sigma}(\tau) = \mathrm{e}^{\tau H} c_{\boldsymbol{k}\sigma} \mathrm{e}^{-\tau H}, \tag{11}$$

the matrix element becomes

$$\langle \{n\} | c_{\boldsymbol{k}\sigma}(\tau) | \{m\} \rangle = \mathrm{e}^{\tau\left(E_{\{n\}} - E_{\{m\}}\right)} \langle \{n\} | c_{\boldsymbol{k}\sigma} | \{m\} \rangle \tag{12}$$

Therefore

$$G_\sigma(\boldsymbol{k}, \tau) = -\frac{1}{Z} \sum_{\{n\},\{m\}} \mathrm{e}^{-\beta E_{\{n\}}} \mathrm{e}^{\tau\left(E_{\{n\}} - E_{\{m\}}\right)} \langle \{n\} | c_{\boldsymbol{k}\sigma} | \{m\} \rangle \left\langle \{m\} \middle| c^\dagger_{\boldsymbol{k}\sigma} \middle| \{n\} \right\rangle. \tag{13}$$

Now the matrix elements are only nonzero if

$$n_{\boldsymbol{k}\sigma} = 0 \quad \text{and} \quad m_{\boldsymbol{k}\sigma} = 1 \quad \text{and } n_{\boldsymbol{k}'\sigma'} = m_{\boldsymbol{k}'\sigma'} \text{ for all } \boldsymbol{k}'\sigma' \neq \boldsymbol{k}\sigma \tag{14}$$

so we can conclude

$$E_{\{m\}} = E_{\{n\}} + \xi_{\boldsymbol{k}} \tag{15}$$

and the product of matrix elements gives 1, so

$$G_\sigma(\boldsymbol{k}, \tau) = -\frac{1}{Z} \sum_{\{n\}, n_{\boldsymbol{k}\sigma}=0} \mathrm{e}^{-\beta E_{\{n\}}} \mathrm{e}^{-\tau \xi_{\boldsymbol{k}}}. \tag{16}$$

We can perform the sum

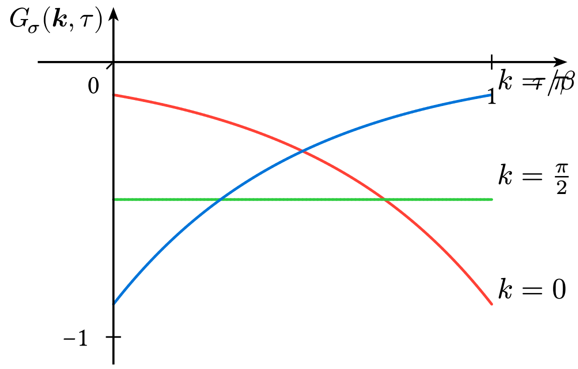
$$\sum_{\{n\},n_{\boldsymbol{k}\sigma}=0} \mathrm{e}^{-\beta E_{\{n\}}} = \prod_{\boldsymbol{k}'\sigma'\neq\boldsymbol{k}\sigma} \sum_{n_{\boldsymbol{k}'\sigma'}=0}^{1} \mathrm{e}^{-\beta\xi_{\boldsymbol{k}'}n_{\boldsymbol{k}'\sigma'}} = \prod_{\boldsymbol{k}'\sigma'\neq\boldsymbol{k}\sigma} \left(1+\mathrm{e}^{-\beta\xi_{\boldsymbol{k}'}}\right) \tag{17}$$

and then get

$$G_\sigma(\boldsymbol{k},\tau) = -\mathrm{e}^{-\tau\xi_{\boldsymbol{k}}} \frac{\prod_{\boldsymbol{k}'\sigma'\neq\boldsymbol{k}\sigma}\left(1+\mathrm{e}^{-\beta\xi_{\boldsymbol{k}'}}\right)}{\prod_{\boldsymbol{k}'\sigma'}\left(1+\mathrm{e}^{-\beta\xi_{\boldsymbol{k}'}}\right)} \tag{18}$$

and are able to cancel common factors

$$G_\sigma(\boldsymbol{k},\tau) = -\frac{\mathrm{e}^{-\tau\xi_{\boldsymbol{k}}}}{1+\mathrm{e}^{-\beta\xi_{\boldsymbol{k}}}} = -\mathrm{e}^{-\tau\xi_{\boldsymbol{k}}}[1-n_{\mathrm{F}}(\xi_{\boldsymbol{k}})]. \tag{19}$$



b)

PROBLEM

Continue the result obtained above for $G_\sigma(\boldsymbol{k},\tau)$ to real times by the inverse Wick-rotation $\tau \to \mathrm{i}t$. Give a physical interpretation for the result.

SOLUTION

From the imaginary-time result for $0 < \tau < \beta$,

$$G_\sigma(\boldsymbol{k},\tau) = -\frac{\mathrm{e}^{-\tau\xi_{\boldsymbol{k}}}}{1+\mathrm{e}^{-\beta\xi_{\boldsymbol{k}}}} = -\mathrm{e}^{-\tau\xi_{\boldsymbol{k}}}[1-n_{\mathrm{F}}(\xi_{\boldsymbol{k}})]. \tag{20}$$

Performing the inverse Wick rotation $\tau \to it$ gives

$$G_\sigma(\boldsymbol{k},t) = -\frac{\mathrm{e}^{-\mathrm{i}t\xi_{\boldsymbol{k}}}}{1+\mathrm{e}^{-\beta\xi_{\boldsymbol{k}}}} = -\mathrm{e}^{-\mathrm{i}t\xi_{\boldsymbol{k}}}[1-n_{\mathrm{F}}(\xi_{\boldsymbol{k}})]. \tag{21}$$

This is the continuation of the positive imaginary-time branch. It corresponds to the propagation of an added particle in the state $(\boldsymbol{k},\sigma)$. The phase factor $\mathrm{e}^{-\mathrm{i}t\xi_{\boldsymbol{k}}}$ describes free time evolution with energy measured relative to the chemical potential, while the factor $1 - n_{\mathrm{F}}(\xi_{\boldsymbol{k}})$ is the probability that the state is initially empty and can therefore accept an added fermion.

c)

PROBLEM

Calculate the Green's function $G_\sigma(\boldsymbol{k}, \mathrm{i}\omega_n)$ in Matsubara frequency space by performing the Fourier-transform

$$G_\sigma(\boldsymbol{k}, \mathrm{i}\omega_n) = \int_0^\beta \mathrm{d}\tau \mathrm{e}^{\mathrm{i}\omega_n \tau} G_\sigma(\boldsymbol{k}, \tau), \tag{22}$$

where $\omega_n = \frac{\pi}{\beta}(2n+1)$, $n \in \mathbb{Z}$ is a fermionic Matsubara frequency. Then continue the results on the real frequency axis and calculate the corresponding spectral function $A(\boldsymbol{k}, \omega)$.

SOLUTION

We perform the Fourier transform

$$G_\sigma(\boldsymbol{k}, \mathrm{i}\omega_n) = \int_0^\beta \mathrm{d}\tau \mathrm{e}^{\mathrm{i}\omega_n \tau} G_\sigma(\boldsymbol{k}, \tau), \tag{23}$$

inserting the result from above

$$G_\sigma(\boldsymbol{k}, \mathrm{i}\omega_n) = -[1 - n_\mathrm{F}(\xi_{\boldsymbol{k}})] \int_0^\beta \mathrm{d}\tau \mathrm{e}^{\mathrm{i}\omega_n \tau} \mathrm{e}^{-\tau \xi_{\boldsymbol{k}}} \tag{24}$$

to get

$$G_\sigma(\boldsymbol{k}, \mathrm{i}\omega_n) = \frac{1}{\mathrm{i}\omega_n - \xi_{\boldsymbol{k}}}. \tag{25}$$

More explicitly,

$$G_\sigma(\boldsymbol{k}, \mathrm{i}\omega_n) = -[1 - n_\mathrm{F}(\xi_{\boldsymbol{k}})] \int_0^\beta \mathrm{d}\tau \mathrm{e}^{(\mathrm{i}\omega_n - \xi_{\boldsymbol{k}})\tau}. \tag{26}$$

Hence

$$G_\sigma(\boldsymbol{k}, \mathrm{i}\omega_n) = -[1 - n_\mathrm{F}(\xi_{\boldsymbol{k}})] \frac{\mathrm{e}^{(\mathrm{i}\omega_n - \xi_{\boldsymbol{k}})\beta} - 1}{\mathrm{i}\omega_n - \xi_{\boldsymbol{k}}}. \tag{27}$$

Since ω_n is fermionic,

$$\mathrm{e}^{\mathrm{i}\omega_n \beta} = -1, \tag{28}$$

and therefore

$$\mathrm{e}^{(\mathrm{i}\omega_n - \xi_{\boldsymbol{k}})\beta} - 1 = -\left(1 + \mathrm{e}^{-\beta \xi_{\boldsymbol{k}}}\right). \tag{29}$$

Using

$$1 - n_\mathrm{F}(\xi_{\boldsymbol{k}}) = \frac{1}{1 + \mathrm{e}^{-\beta \xi_{\boldsymbol{k}}}}, \tag{30}$$

the thermal factors cancel and one obtains

$$G_\sigma(\boldsymbol{k}, \mathrm{i}\omega_n) = \frac{1}{\mathrm{i}\omega_n - \xi_{\boldsymbol{k}}}. \tag{31}$$

To continue the results, we set $\mathrm{i}\omega_n \to \omega + \mathrm{i}0^+$ and get

$$G_\sigma(\boldsymbol{k},\omega) = \frac{1}{\omega - \xi_{\boldsymbol{k}} + \mathrm{i}0^+} \tag{32}$$

as well as the spectral function

$$A(\boldsymbol{k},\omega) = -\frac{1}{\pi}\operatorname{Im} G_\sigma(\boldsymbol{k},\omega) = \delta(\omega - \xi_{\boldsymbol{k}}). \tag{33}$$

d)

PROBLEM

Now, consider the opposite limit where the kinetic energy appearing in the Hamiltonian in Equation 1 is negligible compared to the interaction, i.e., $\varepsilon_{\boldsymbol{k}} = 0$ (atomic limit).

SOLUTION

In the atomic limit, the Hamiltonian is reduced to

$$H = \sum_i H_i \tag{34}$$

with

$$H_i = U n_{i\uparrow} n_{i\downarrow} - \mu\big(n_{i\uparrow} + n_{i\downarrow}\big) \tag{35}$$

and we work at half filling $\mu = \frac{U}{2}$.

So it suffices to work in the local site i (the label of which we'll omit), where we write down the Fock basis

$$|0\rangle \qquad |\uparrow\rangle \qquad |\downarrow\rangle \qquad |\uparrow\downarrow\rangle = c_\uparrow^\dagger c_\downarrow^\dagger |0\rangle \tag{36}$$

which is already an eigenbasis with eigenvalues

$$0 \qquad -\frac{U}{2} \qquad -\frac{U}{2} \qquad 0. \tag{37}$$

Since the atoms are independent, the full problem factorizes into identical single-site problems.

e)

PROBLEM

The local Green's function for site i, $G_{i\sigma}(\tau)$, is defined as

$$G_{i\sigma}(\tau) = -\left\langle T_\tau \hat{c}_{i\sigma}(\tau)\hat{c}_{i\sigma}^\dagger\right\rangle = -\frac{1}{Z}\operatorname{tr}\left[\mathrm{e}^{-\beta\hat{H}}\hat{c}_{i\sigma}(\tau)\hat{c}_{i\sigma}^\dagger\right], \quad \beta \geq \tau \geq 0. \tag{38}$$

Note that the different atoms are completely independent in this case and the local Green's function is thus the same as already calculated in Problem 6 of Exercise 3. From its Fourier transform $G_{i\sigma}(\mathrm{i}\omega_n)$, extract the corresponding expression for the self-energy $\Sigma_{i\sigma}(\mathrm{i}\omega_n)$. Is the atomic-limit expression derivable within conventional perturbation theory?

We first compute the partition function

$$Z = \sum_n \langle n|\mathrm{e}^{-\beta H}|n\rangle = 2\big(1+\mathrm{e}^{\beta U/2}\big) \tag{39}$$

and then the Green's function. To that end, we start from Equation 13, which we restate here in adapted form for convenience

$$G_\sigma(\tau) = -\frac{1}{Z}\sum_{n,m} \mathrm{e}^{-\beta E_n}\mathrm{e}^{\tau(E_n-E_m)}\langle n|c_\sigma|m\rangle\langle m|c_\sigma^\dagger|n\rangle. \tag{40}$$

and compute the nonzero matrix-elements

$$\begin{aligned}
\langle 0|c_\sigma|\sigma\rangle &= 1 & \langle \sigma|c_\sigma^\dagger|0\rangle &= 1\\
\langle \downarrow|c_\uparrow|\uparrow\downarrow\rangle &= 1 & \langle \uparrow\downarrow|c_\uparrow^\dagger|\downarrow\rangle &= 1\\
\langle \uparrow|c_\downarrow|\uparrow\downarrow\rangle &= -1 & \langle \uparrow\downarrow|c_\downarrow^\dagger|\uparrow\rangle &= -1,
\end{aligned} \tag{41}$$

implying for either spin

$$G_\sigma(\tau) = -\frac{1}{Z}\big(\mathrm{e}^{\tau U/2} + \mathrm{e}^{\beta U/2}\mathrm{e}^{-\tau U/2}\big) = -\frac{1}{2}\frac{\mathrm{e}^{\tau U/2}+\mathrm{e}^{(\beta-\tau)U/2}}{1+\mathrm{e}^{\beta U/2}}. \tag{42}$$

and defining

$$f = n_\mathrm{F}\left(\frac{U}{2}\right) = \big[1+\mathrm{e}^{\beta U/2}\big]^{-1} \tag{43}$$

we can write

$$G_\sigma(\tau) = -\frac{1}{2}\big[f\mathrm{e}^{\tau U/2} + (1-f)\mathrm{e}^{-\tau U/2}\big]. \tag{44}$$

Fourier transforming yields

$$G_\sigma(\mathrm{i}\omega_n) = \frac{1}{2}\left[\frac{1}{\mathrm{i}\omega_n+\frac{U}{2}} + \frac{1}{\mathrm{i}\omega_n-\frac{U}{2}}\right] = \frac{\mathrm{i}\omega_n}{(\mathrm{i}\omega_n)^2-\frac{U^2}{4}} \tag{45}$$

and thus a self-energy of

$$\Sigma_\sigma(\mathrm{i}\omega_n) = \frac{1}{G_{0\sigma}(\mathrm{i}\omega_n)} - \frac{1}{G_\sigma(\mathrm{i}\omega_n)} = \frac{U}{2} + \frac{U^2}{4\mathrm{i}\omega_n}. \tag{46}$$

because

$$G_{0\sigma}(\mathrm{i}\omega_n) = \frac{1}{\mathrm{i}\omega_n+\mu}. \tag{47}$$

The self-energy contains a pole at zero Matsubara frequency,

$$\Sigma_\sigma(\mathrm{i}\omega_n) = \frac{U}{2} + \frac{U^2}{4\mathrm{i}\omega_n}. \tag{48}$$

This pole is the atomic-limit signature of the splitting into lower and upper Hubbard levels. It is singular at $\mathrm{i}\omega_n = 0$ and therefore cannot be obtained from conventional regular perturbation theory around the non-interacting limit. In ordinary weak-coupling perturbation theory one expands in powers of U at fixed non-interacting propagator, whereas the atomic limit is controlled by the local interaction and already contains the non-perturbative separation of the spectrum into two Hubbard peaks.

f)

PROBLEM

Calculate, analogously as above, the local magnetic (spin) $\chi_i^{\mathrm{s}}(\tau) = \langle T_\tau S_i^z(\tau) S_i^z \rangle$ and density (charge) $\chi_i^{\mathrm{c}}(\tau) = \langle T_\tau n_i(\tau) n_i \rangle$ susceptibilities in the atomic limit of the Hubbard model (with $S_i^z = n_{i\uparrow} - n_{i\downarrow}$ and $n_i = n_{i\uparrow} + n_{i\downarrow}$), as well as their Fourier transform to Matsubara frequencies. Then, analytically continue the expressions to real frequencies. What can you say about the temperature dependence?

SOLUTION

We observe, again omitting the site label i, that both spin and charge operators commute with the Hamiltonian

$$[S^z, H] = 0 = [n, H] \tag{49}$$

because $[n_\sigma, n_{\sigma'}] = 0$ and so both susceptibilities lose their imaginary time dependence and become simple thermal averages

$$\begin{aligned} \chi^{\mathrm{s}} &= \left\langle (S^z)^2 \right\rangle \\ \chi^{\mathrm{c}} &= \langle n^2 \rangle . \end{aligned} \tag{50}$$

We start by writing for the local magnetic (spin) susceptibility

$$\chi^{\mathrm{s}} = \frac{1}{Z} \sum_m \mathrm{e}^{-\beta E_m} \langle m | S^z S^z | m \rangle \tag{51}$$

and expand out

$$(S^z)^2 = n_\uparrow^2 - 2 n_\uparrow n_\downarrow + n_\downarrow^2 \tag{52}$$

to find

$$\begin{aligned} \left\langle 0 \middle| (S^z)^2 \middle| 0 \right\rangle &= 0 \\ \left\langle \uparrow \middle| (S^z)^2 \middle| \uparrow \right\rangle &= 1 \\ \left\langle \downarrow \middle| (S^z)^2 \middle| \downarrow \right\rangle &= 1 \\ \left\langle \uparrow\downarrow \middle| (S^z)^2 \middle| \uparrow\downarrow \right\rangle &= 0 \end{aligned} \tag{53}$$

and thus

$$\chi^{\mathrm{s}} = \frac{1}{Z} 2 \mathrm{e}^{\beta U/2} = \frac{\mathrm{e}^{\beta U/2}}{1 + \mathrm{e}^{\beta U/2}} = 1 - f \tag{54}$$

with

$$f \equiv n_{\mathrm{F}} \left(\frac{U}{2} \right). \tag{55}$$

The local density (charge) susceptibility, on the other hand, is given by

$$\chi^{\mathrm{c}} = \frac{1}{Z} \sum_m \mathrm{e}^{-\beta E_m} \langle m|nn|m\rangle \tag{56}$$

and we follow the same path as before

$$n^2 = n_\uparrow^2 + 2n_\uparrow n_\downarrow + n_\downarrow^2 \tag{57}$$

$$\begin{aligned} \langle 0|n^2|0\rangle &= 0 \\ \langle\uparrow|n^2|\uparrow\rangle &= 1 \\ \langle\downarrow|n^2|\downarrow\rangle &= 1 \\ \langle\uparrow\downarrow|n^2|\uparrow\downarrow\rangle &= 4 \end{aligned} \tag{58}$$

to get

$$\chi^{\mathrm{c}} = \frac{1}{Z}\left(4 + 2\mathrm{e}^{\beta U/2}\right) = \frac{2 + \mathrm{e}^{\beta U/2}}{1 + \mathrm{e}^{\beta U/2}} = 1 + f. \tag{59}$$

The Fourier transforms are then easy to compute

$$\begin{aligned} \chi^{\mathrm{s}}(\mathrm{i}\omega_n) &= \beta(1-f)\delta_{n,0} \\ \chi^{\mathrm{c}}(\mathrm{i}\omega_n) &= \beta(1+f)\delta_{n,0} \end{aligned} \tag{60}$$

and continue

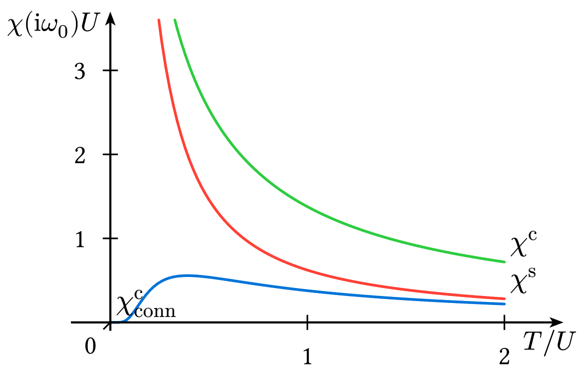
$$\begin{aligned} \chi^{\mathrm{s}}(\omega) &= (1-f)\delta(\omega) \\ \chi^{\mathrm{c}}(\omega) &= (1+f)\delta(\omega). \end{aligned} \tag{61}$$

We can also compute the connected charge susceptibility

$$\chi^{\mathrm{c}}_{\mathrm{conn}}(\tau) \equiv \langle T_\tau n(\tau) n(0)\rangle - \langle n\rangle^2 = f \tag{62}$$

and transform to Matsubara

$$\chi^{\mathrm{c}}_{\mathrm{conn}}(\mathrm{i}\omega_n) = \beta f \delta_{n,0}. \tag{63}$$



SECTION

013

RPA for the Hubbard model

In the Hubbard model (Hamiltonian in Equation 1), the interaction is purely local and penalizes double occupations: $U\sum_i n_{i\uparrow}n_{i\downarrow}$. Therefore, the interaction only couples electrons with opposite spin.

Then also susceptibilities can acquire a spin-dependence: $\chi_{\sigma\sigma'}$.

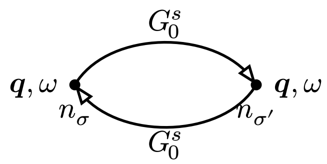
Remembering that momentum, energy and spin need to be conserved at each vertex:

a)

PROBLEM

Draw the bubble diagram of the free susceptibility $\chi_0^{\sigma\sigma'}(\boldsymbol{q},\omega)$ and say which spin-combinations are possible.

SOLUTION



The free susceptibility is diagonal in spin:

$$\chi_0^{\sigma\sigma'}(\boldsymbol{q},\omega) = \delta_{\sigma\sigma'}\chi_0^{\sigma(\boldsymbol{q},\omega)}. \tag{64}$$

This is because the density vertices conserve spin, and the free propagators G_0 do not flip spin. Therefore the spin running around the bubble must be the same on both sides.

Hence the only nonzero components are

$$\chi_0^{\uparrow\uparrow} \quad \text{and} \quad \chi_0^{\downarrow\downarrow}, \tag{65}$$

while

$$\chi_0^{\uparrow\downarrow} = \chi_0^{\downarrow\uparrow} = 0. \tag{66}$$

b)

PROBLEM

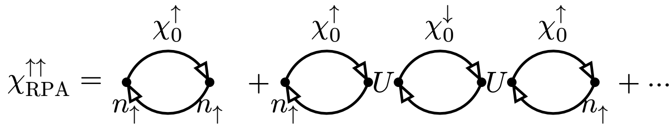
Draw the random phase approximation (RPA) series for $\chi_{\mathrm{RPA}}^{\uparrow\uparrow}$ and $\chi_{\mathrm{RPA}}^{\uparrow\downarrow}$. What can you say about the allowed powers of U in both series? Translate the diagrams into formulas and rewrite them using the geometric series. In all of this you can omit the labels for momentum and frequency.

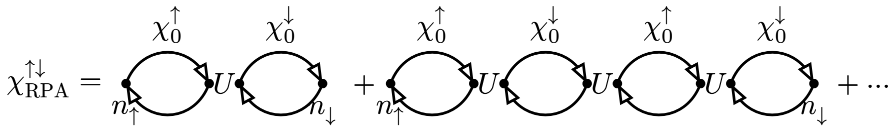
SOLUTION

The Hubbard interaction has the form

$$U n_\uparrow n_\downarrow, \tag{67}$$

so every interaction vertex couples an $\uparrow$ density bubble to an $\downarrow$ density bubble. Thus the RPA chain alternates spin sectors.





The same-spin response contains an even number of Hubbard vertices:

$$\chi^{\uparrow\uparrow}_{\mathrm{RPA}} = \chi_0^\uparrow + \chi_0^\uparrow U \chi_0^\downarrow U \chi_0^\uparrow + \cdots \tag{68}$$

The opposite-spin response contains an odd number of Hubbard vertices:

$$\chi^{\uparrow\downarrow}_{\mathrm{RPA}} = -\chi_0^\uparrow U \chi_0^\downarrow - \chi_0^\uparrow U \chi_0^\downarrow U \chi_0^\uparrow U \chi_0^\downarrow - \cdots \tag{69}$$

We observe

$$\begin{aligned} \chi^{\uparrow\uparrow}_{\mathrm{RPA}} &= \chi_0^\uparrow \left[1 - U \chi^{\downarrow\uparrow}_{\mathrm{RPA}}\right] \\ \chi^{\uparrow\downarrow}_{\mathrm{RPA}} &= -\chi_0^\uparrow U \chi^{\downarrow\downarrow}_{\mathrm{RPA}}, \end{aligned} \tag{70}$$

and plug the second equation into the first

$$\chi^{\uparrow\uparrow}_{\mathrm{RPA}} = \chi_0^\uparrow \left[1 + U \chi_0^\downarrow U \chi^{\uparrow\uparrow}_{\mathrm{RPA}}\right] \tag{71}$$

and solve to

$$\chi^{\uparrow\uparrow}_{\mathrm{RPA}} = \frac{\chi_0^\uparrow}{1 - U^2 \chi_0^\uparrow \chi_0^\downarrow}. \tag{72}$$

For the other spin channel, we get

$$\chi^{\uparrow\downarrow}_{\mathrm{RPA}} = -\frac{U \chi_0^\uparrow \chi_0^\downarrow}{1 - U^2 \chi_0^\uparrow \chi_0^\downarrow}. \tag{73}$$

c)

PROBLEM

The charge and spin susceptibilities (the local versions of which were already introduced above) are given by

$$\chi^{\mathrm{c}} = \chi^{\uparrow\uparrow} + \chi^{\uparrow\downarrow}, \qquad \chi^{\mathrm{s}} = \chi^{\uparrow\uparrow} - \chi^{\uparrow\downarrow}. \tag{74}$$

Using the result from above give expressions for these susceptibilities in the RPA. Which of the two χs was discussed in the lecture in the context of screening?

SOLUTION

Here, we will assume SU(2) symmetry, which implies

$$\chi_0^{\downarrow} = \chi_0^{\uparrow} = \chi_0 \tag{75}$$

and then RPA yields for the density susceptibility

$$\begin{aligned}\chi^{\mathrm{c}} &= \chi^{\uparrow\uparrow} + \chi^{\uparrow\downarrow} \\ &= \frac{\chi_0}{1-U^2\chi_0^2} - \frac{U\chi_0^2}{1-U^2\chi_0^2} \\ &= \frac{\chi_0(1-U\chi_0)}{(1-U\chi_0)(1+U\chi_0)} \\ &= \frac{\chi_0}{1+U\chi_0}\end{aligned} \tag{76}$$

and for the spin susceptibility

$$\begin{aligned}\chi^{\mathrm{s}} &= \chi^{\uparrow\uparrow} - \chi^{\uparrow\downarrow} \\ &= \frac{\chi_0}{1-U^2\chi_0^2} + \frac{U\chi_0^2}{1-U^2\chi_0^2} \\ &= \frac{\chi_0(1+U\chi_0)}{(1-U\chi_0)(1+U\chi_0)} \\ &= \frac{\chi_0}{1-U\chi_0}.\end{aligned} \tag{77}$$

In the lecture, to derive Thomas-Fermi screening, we used

$$V_{\mathrm{eff}} = \frac{V}{1+V\chi_0}, \tag{78}$$

which corresponds to χ^{c}.

d)

PROBLEM •

(Bonus points) Using the results of Problem 11 of Exercise 5, consider the electronic system for $d = 2$ in presence of the Hubbard interaction $U > 0$, and calculate within the RPA the two (ferromagnetic and antiferromagnetic) spin susceptibilities. On the basis of your RPA calculations, make your final considerations about the tendency of the system towards a given magnetic order at $T = 0$.

SOLUTION

In Problem 11 we found for the non-interacting ferro- and antiferromagnetic susceptibilities of the hypercubic lattice

$$\begin{aligned}\chi_0^{\mathrm{F}} &= \frac{\hbar^2}{4}\Pi_0^{\mathrm{F}} \\ \chi_0^{\mathrm{AF}} &= \frac{\hbar^2}{4}\Pi_0^{\mathrm{AF}}\end{aligned} \tag{79}$$

with the bare polarization bubbles

$$\begin{aligned}\Pi_0^{\mathrm{F}} &= \frac{1}{N}\sum_{\boldsymbol{k}} \frac{\beta}{2}\left[\cosh\left(\frac{\beta\varepsilon_{\boldsymbol{k}}}{2}\right)\right]^{-2} \\ \Pi_0^{\mathrm{AF}} &= \frac{1}{N}\sum_{\boldsymbol{k}} \frac{1}{\varepsilon_{\boldsymbol{k}}}\tanh\left(\frac{\beta\varepsilon_{\boldsymbol{k}}}{2}\right).\end{aligned} \tag{80}$$

We now add an interaction

$$H_{\mathrm{int}} = U n_{\downarrow} n_{\uparrow} \tag{81}$$

to the Hamiltonian and approximate the susceptibilities in RPA

$$\begin{aligned}\chi_{\mathrm{RPA}}^{\mathrm{F}} &= \frac{\hbar^2}{4}\frac{\Pi_0^{\mathrm{F}}}{1 - U\Pi_0^{\mathrm{F}}} \\ \chi_{\mathrm{RPA}}^{\mathrm{AF}} &= \frac{\hbar^2}{4}\frac{\Pi_0^{\mathrm{AF}}}{1 - U\Pi_0^{\mathrm{AF}}}.\end{aligned} \tag{82}$$

Ferromagnetic
Taking the $T \to 0$ (i.e. $\beta \to \infty$) limit, we note that

$$\lim_{\beta\to\infty} \frac{\beta}{2}\left[\cosh\left(\frac{\beta\varepsilon}{2}\right)\right]^{-2} = 2\delta(\varepsilon) \tag{83}$$

and therefore

$$\Pi_0^{\mathrm{F}} = \frac{2}{N}\sum_{\boldsymbol{k}} \delta(\varepsilon_{\boldsymbol{k}}) = 2N(0), \tag{84}$$

where $N(0)$ is the DOS at the Fermi level.

Antiferromagnetic
Because

$$\lim_{\beta\to\infty} \frac{1}{\varepsilon}\tanh\left(\frac{\beta\varepsilon}{2}\right) = \frac{1}{|\varepsilon|}, \tag{85}$$

we have at $T \to 0$

$$\Pi_0^{\mathrm{AF}} = \frac{1}{N}\sum_{\boldsymbol{k}} \frac{1}{|\varepsilon_{\boldsymbol{k}}|} \tag{86}$$

In two dimensions at half filling the square lattice is perfectly nested. The nesting vector is

$$\boldsymbol{Q} = (\pi, \pi), \tag{87}$$

and it satisfies

$$\varepsilon_{\boldsymbol{k}+\boldsymbol{Q}} = -\varepsilon_{\boldsymbol{k}}. \tag{88}$$

Therefore the antiferromagnetic particle-hole bubble diverges at $T = 0$. In RPA this means that

$$\chi_{\mathrm{RPA}}^{\mathrm{AF}} = \frac{\hbar^2}{4}\frac{\Pi_0^{\mathrm{AF}}}{1 - U\Pi_0^{\mathrm{AF}}} \tag{89}$$

becomes unstable when

$$1 - U\Pi_0^{\mathrm{AF}} = 0. \tag{90}$$

Since Π_0^{AF} diverges for $T \to 0$, this instability occurs for arbitrarily small repulsive $U > 0$.

The ferromagnetic susceptibility is also enhanced because the two-dimensional square lattice has a van-Hove singularity at half filling. However, the antiferromagnetic channel is the dominant one because it is additionally enhanced by perfect nesting.

Thus, within RPA, the half-filled two-dimensional Hubbard model has a dominant tendency towards antiferromagnetic order at $T = 0$.